# Real-Time Biofeedback Metrics for Couples Therapy

Simplified HR / HRV-only reference

## Contents

**Real-Time Biofeedback Metrics for Couples Therapy** **1**
Input requirement . . . . . . . . . . 1
The six core metrics . . . . . . . . . . 2
The one tradeoff to keep in mind . . . . . . . . . . 4
Personalized baselining . . . . . . . . . . 4
Real-time cadence . . . . . . . . . . 4
Validation and limitations . . . . . . . . . . 5
Key sources . . . . . . . . . . 5

## Real-Time Biofeedback Metrics for Couples Therapy

### Simplified HR / HRV-only reference

This is a streamlined metric set for a real-time couples-therapy biofeedback system that uses **only heart rate and heart rate variability** — no dedicated respiration belt. Respiration is *inferred from the heartbeat series itself*, which is sufficient for everything this system needs.

The key idea: breathing already shows up in HRV as the **high-frequency oscillation** of the RR interval series. So removing the respiration belt does not remove the respiratory dimension — it just accesses it through the tachogram. Only two things from a full design genuinely need a separate belt (peak-to-trough RSA and respiration–cardiac cross-coherence), and both have clean HR/HRV-only replacements.

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### Input requirement

All metrics are computed from **beat-to-beat RR intervals** (interbeat intervals, in milliseconds) — the time between successive heartbeats from ECG R-peaks or PPG systolic peaks. This is **not** an averaged heart-rate-in-bpm stream; a smoothed bpm number has discarded the beat-to-beat variation that the metrics depend on.

**Preprocessing.** Reject artifacts first: keep intervals in 300–2000 ms (30–200 bpm), and drop any interval differing from its neighbours' running median by more than

~20% (ectopic/misdetection filter). Every metric below inherits any error left in the RR series.

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## The six core metrics

In priority order. The first two are load-bearing; the rest add depth.

### 1. Mean HR and the flooding flag

The most clinically actionable signal, and the simplest. Gottman's *Diffuse Physiological Arousal* (flooding) is the state where heart rate climbs to the point that empathy and listening go offline — classically near 100 bpm, but the threshold is individual, so express it against the person's own baseline.

$$\text{mean HR} = \frac{60000}{\overline{RR}} \quad [\text{bpm}]$$

Flag flooding when HR exceeds **either** an absolute ceiling (~95–100 bpm) **or** the personal baseline by ~10% (some use 15–20%), *sustained* for several seconds:

$$\text{flooded} = (\text{HR} \geq 1.10 \times \text{HR}_{\text{baseline}}) \ \text{ OR } \ (\text{HR} \geq 95)$$

Pair the flag with its intervention: a ~20-minute break with active distraction, since the stress hormones driving the state clear slowly.

### 2. RMSSD

The short-window workhorse: vagally mediated, drops reliably under stress, and stays valid on the brief (10–30 s) windows real-time feedback needs. With $d_i = RR_{i+1} - RR_i$:

$$\text{RMSSD} = \sqrt{\frac{1}{N-1}\sum_i d_i^2} \quad [\text{ms}]$$

When a partner activates, expect RMSSD to fall.

### 3. HF power — the RSA / vagal-tone proxy

This replaces belt-based peak-to-trough RSA. HF-band power is the standard frequency-domain stand-in for respiratory sinus arrhythmia and vagal tone, and it is computed **purely from RR intervals**:

$$P_{HF} = \int_{0.15}^{0.40} S(f)\, df$$

where $S(f)$ is the power spectrum of the RR series. It indexes the same construct as RSA — parasympathetic / emotion-regulation capacity — and falls with activation. Higher HF tracks better regulation; a drop tracks vagal withdrawal.

**Spectral estimation on short windows:** use the **Lomb–Scargle periodogram**, which works directly on the unevenly sampled, event-timed RR series and avoids interpolation artifacts. Switch to a Welch estimate on a 4 Hz resampled tachogram only once you have ≥60 s of clean data.

### 4. Derived respiration rate (free, from the tachogram)

Because breathing *is* the HF oscillation, respiration rate comes for free as the frequency of the dominant HF peak:

$$\text{resp rate} = 60 \times f_{\text{peak}} \ \text{[breaths/min]}, \qquad f_{\text{peak}} = \arg\max_{0.15 \le f \le 0.40} S(f)$$

It is a useful arousal proxy on its own (resting adults breathe ~9–20/min, and rate rises with activation) and — importantly — it doubles as the caveat flag described below.

### 5. Coherence — the biofeedback *target*

Coherence is the regulated state to *move toward*, roughly the inverse of flooding. It measures how concentrated the heart-rate oscillation is around the ~0.1 Hz resonance frequency reached by slow breathing (~6 breaths/min). From the HRV spectrum (McCraty method):

1. Find the highest peak $f_0$ in 0.04–0.26 Hz.
2. Integrate power in a 0.030 Hz window centered on $f_0$ (the *peak power*).
3. Take the ratio to the rest of the spectrum:

$$\text{Coherence} = \frac{P_{\text{peak}}}{P_{\text{total}} - P_{\text{peak}}}$$

A single dominant ~0.1 Hz oscillation gives a high score. This is the natural thing to display to a partner during a self-soothing break — and it needs only the RR series.

### 6. Dyadic coupling — lagged cross-correlation

The relational layer, and it never needed respiration. Put both partners' instantaneous HR (or their RMSSD / HF time series) on a common, evenly sampled grid, then slide one against the other across a range of lags and take the strongest correlation:

$$\text{peak\_r} = \max_{|\tau| \le \tau_{\max}} |\text{corr}(A_t,\ B_{t+\tau})|$$

The lag $\tau$ at the maximum indicates which partner leads. Four cautions:

- **Detrend each window first** — shared slow drift fakes synchrony.
- **Sign matters** — positive is in-phase, negative is anti-phase; they mean different things.
- **Use a surrogate baseline** — compare against shuffled / non-partner pairings before calling coupling meaningful.
- **Interpret jointly with arousal** — high synchrony while both are flooded means "locked in conflict"; high synchrony while both are coherent means co-regulation. Treat per-partner arousal/coherence as the reliable core and synchrony as the exploratory overlay.

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## The one tradeoff to keep in mind

HF power is a slightly noisier RSA stand-in than the belt-based version, because without a breathing waveform you cannot correct it for respiration rate and tidal volume. Two practical consequences:

- **Pair HF with the derived respiration rate**, so you can tell when a change in HF is really just a change in breathing rather than a change in vagal tone.
- **During slow-breathing / self-soothing exercises, watch coherence and the ~0.1 Hz peak — not HF.** Slow breathing (below ~9/min, i.e. 0.15 Hz) pushes the respiratory peak *down out of the HF band and into the LF band*, so HF will under-read exactly when the person is calming down. Use **HF for spontaneous activation, coherence for the regulation exercise.**

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## Personalized baselining

Every threshold above is individual. At the start of a session, record a few minutes of calm baseline per partner, store the mean and standard deviation of each metric, and express live values as z-scores or percent change against it:

$$z = \frac{x - \mu_{\text{baseline}}}{\sigma_{\text{baseline}}}$$

This is also what makes a composite score meaningful: a convergent **activation index** combines the channels that should move together — HR and respiration rate up, RMSSD and HF and coherence down — each in baseline z-units. Convergence across channels is far more trustworthy than any single one spiking alone.

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## Real-time cadence

Different metrics need different amounts of data, so run them at different update rates from a rolling buffer:

| Tier | Window | Metrics | Why |
|---|---|---|---|
| Fast | per beat | mean HR | Immediate display, minimal data |
| Mid | ~30 s rolling | RMSSD, flooding flag | RMSSD stays valid at short lengths |
| Slow | ~60 s rolling | HF, respiration rate, coherence | Need several oscillation cycles |
| Dyadic | longer + lag search | synchrony | Computed at the dyad level |

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## Validation and limitations

- These are **physiological proxies, not emotion readouts** — validate every metric and threshold against self-report and/or therapist coding before trusting it in session.
- **Confirm your sensor exposes beat-to-beat intervals**, not a smoothed bpm number, or most of this cannot be computed as written.
- **Tunable parameters to fix empirically:** the flooding margin and sustain duration, the synchrony window length and maximum lag, the activation-index weights, and the artifact-rejection tolerance.
- The dyadic synchrony literature is genuinely heterogeneous — keep synchrony in the exploratory layer.

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## Key sources

- **Levenson & Gottman (1983, 1985)** — physiological linkage and the flooding / DPA framework.
- **Shaffer & Ginsberg; ultra-short-term HRV reviews** — HRV metric definitions, HF power as an RSA/vagal proxy, and short-window validity limits.
- **Lehrer, Vaschillo & Gevirtz; McCraty et al. (2009)** — resonance-frequency breathing (~0.1 Hz) and cardiac coherence.
- **Butler (2015); Palumbo et al. (2017); Timmons et al. (2015)** — reviews of interpersonal physiological synchrony and its context-dependence.